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High dependence on nitrogen-fixing bacteria in a Neotropical legume species

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Abstract
Mutualisms are cooperative interactions between species that exhibit varying degrees of partner dependence. In a greenhouse experiment, we investigated the dependence of the native legume species *Chamaecrista nictitans* (L.) Moench on its nitrogen-fixing bacteria partners, rhizobia. Our study aimed to determine the nature of this association and the level of plant dependence on rhizobia. We conducted a manipulative experiment in a climatically controlled greenhouse using sterilized soil, where we cultivated 180 *C. nictitans* plants, 90 plants with rhizobia (R+), and the remaining 90 plants without rhizobia (R-) inoculation (exclusion treatment). We assessed plant dependence on nitrogen-fixing bacteria by comparing survival rates, growth patterns, and bud and flower production between R+and R- plants. Additionally, we investigated the investment of R+plants in root nodulation and its influence on plant biomass. The results supported a highly dependent association between *C. nictitans* and nitrogen-fixing bacteria. R- plants exhibited increased mortality rates, stunted growth, and a complete absence of reproductive structures. R+plants that produced many nodules had a higher accumulated dry biomass. This study underscores the variability in dependence within mutualistic associations in tropical ecosystems. Understanding partner dependence in these habitats is crucial for unraveling the ecological and evolutionary processes underlying the mutualisms, especially in tropical ecosystems.

Keywords Chamaecrista · Mutualisms · Plant-root association · Rhizobium · Biological nitrogen fixation · Mutualism dependence

1 Introduction

Mutualisms are interactions between species where resources or services are exchanged, resulting in a net benefit for the partners involved (Schemske and Horvitz 1988). However, not all mutualistic interactions have an equal level of dependence. The degree of mutualism dependency is linked to the extent of fitness reduction when partners are not engaged in the interaction (Chomicki et al. 2020). In this way, mutualistic interactions are classified along a continuum ranging from obligatory to facultative based on the level of dependency of the partners involved (Boucher et al. 1982; Bronstein 2015). At the one end of the continuum, obligatory mutualistic interactions occur when at least one partner is highly dependent on the interaction, unable to survive or reproduce without its mutualistic counterpart. Conversely, facultative mutualisms exist at the other extreme, where partners can survive and reproduce independently but achieve enhanced fitness when participating in mutualistic

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interaction (Bronstein 1994). Generally, dependency and specialization are interconnected, meaning species involved in facultative mutualisms have multiple potential mutualistic partners (Bronstein 1994; Stanton 2003; Bascompte and Jordano 2007). Consequently, factors affecting mutualistic partners are anticipated to have a more significant impact on obligate mutualism, as excluding one partner could result in the extinction of the other (Chomicki et al. 2020). Distinguishing mutualistic interactions along the spectrum from obligatory to facultative is crucial, as the level of dependency on mutualistic partners not only molds the evolutionary dynamics of the interaction but also dictates the potential consequences of ruptures or changes in the partnership in a changing world.

In mutualistic interactions between legume plants (Fabaceae) and nitrogen (N)-fixing bacteria, bacteria colonize the plant's root system, forming nodules where atmospheric nitrogen (N_2) is converted into an organic form, primarily ammonia (NH_3), which the plant can utilize (Tate 1995). While the bacteria provide the host plant with N compounds, the plant reciprocates by supplying photosynthetic carbohydrates to the bacteria (Phillips 1980; Tate 1995; Dilworth et al. 2008). This mutualism is an obligatory interaction for some plant species, in which legume plants consistently acquire N from N-fixing bacterial partners (Menge et al. 2009, 2015). However, other legume species can modulate their N uptake rate from N-fixing bacteria, especially in response to soil N availability (Menge et al. 2009, 2015). In this second case, the interaction between legume and bacteria is expected to become more facultative with the higher availability of soil N (Menge et al. 2015). Legume plants facultatively associated with bacteria are widespread among tropical legume plants inhabiting environments characterized by high N richness and availability (Jenny 1950; Hedin et al. 2009). In such tropical settings, legume plants have access to multiple alternative N sources (Sullivan et al. 2014), leading to lower dependency on facultative associations with N-fixing bacteria.

Species belonging to the genus *Chamaecrista* (L.) Moench interacts with N-fixing bacteria by forming root nodules (Sprent et al. 2017). The few studies evaluating the degree of dependency between plants from this genus and N-fixing bacteria indicate that *Chamaecrista* species can occupy different positions along the obligatory-facultative continuum. For instance, *Chamaecrista ramosa* (Vogel) H. S. Irwin & Barneby has a physically integrated relationship with nitrogen-fixing bacteria, indicating the establishment of an obligatory association with N-fixing bacteria (Faria et al. 2022). *Chamaecrista fasciculata* (Michaux), however, has a facultative interaction with these bacteria (Stanton-Geddes and Anderson 2011), thus changing the N absorption from the N-fixing bacteria to the soil N available (Sheffer et al. 2015). Examining the extent to which native legume species rely on N-fixing bacteria is essential for unraveling the ecological and evolutionary dynamics of their interaction, given the significant impact of mutualistic N fixation on plant growth, reproduction, and survival (Batterman et al. 2013; Mathesius 2022). This investigation gains particular importance in scenarios marked by environmental changes, such as alterations in soil N availability (Vitousek et al. 1997). These changes have the potential to profoundly affect the fitness, phenology, and distribution of plant species (Magnoli and Lau 2020). Therefore, understanding the level of dependence becomes a critical factor in anticipating and comprehending the consequences of environmental variations on the overall health and distribution of these legume species.

In this study, we experimentally investigated the degree of dependency of *Chamaecrista nictitans* (L.) Moench, a widespread ruderal legume species in the Americas, on the N-fixing bacteria forming root nodules. The main goal was to assess the relative impact of N-fixing bacteria on *C. nictitans* fitness. To achieve this, we performed a manipulative experiment cultivating plants with and without the presence of N-fixing bacteria under the high availability of soil N. We then measured plant survival, growth, and reproductive output in plants inoculated and non-inoculated with rhizobia under full N availability. Furthermore, we evaluated the investment in nodule production in inoculated plants. This approach allowed us to assess whether the association between *C. nictitans* and its bacteria partners is obligatory or facultative in conditions where N is not a limiting plant resource. More precisely, our assessment aimed to determine: (1) the nature of the mutualistic interaction between *C. nictitans* and rhizobia bacteria, discerning whether it is facultative or obligatory for the plants; (2) the influence of root nodules on *C. nictitans* fitness; and (3) the extent of *C. nictitans* investment in N-fixing root nodules.

2 Materials and methods

2.1 Study system

Chamaecrista nictitans are annual ruderal herbs widespread in the Americas and well distributed in Brazil (Fig. 1a). These herbs have alternate pinnate leaves with touch-sensitive trichomes on the abaxial surface, which triggers leaf closing (Lee 1989). This species is distinguished by the presence of extrafloral nectaries on the upper surface of leaf petioles, and it displays an inducible defensive response system to herbivory (Chinarelli et al. 2022). Inflorescences are arranged in racemes, typically bearing 1–15 yellow flowers with asymmetric morphology. These flowers are self-compatible



Fig. 1 Plants of *Chamaecrista nictitans* (**a**) Adult individual with flowers; (**b**) Root nodules in a *C. nictitans*; (**c**) Three-month-old individuals non-inoculated with rhizobia, and (**d**) inoculated with rhizobia

and able to self-pollinate. Flowering occurs between June and August. Fruits are dry legumes whose opening triggers the ballistic dispersal of seeds (Lee 1989). The root system of this *Chamaecrista* species forms nodules after a signal exchange with rhizobia, creating a hosting environment for the N-fixing rhizobia (Fig. 1b) (Naisbitt, 1992), typically of the genus *Bradyrhizobium* (Parker & Rosteau, 2014; Santos et al., 2017; Casaes et al., 2024). These nodules, observed to contain symbiosomes (Casaes et al. 2023), represent organelles marked by high compartmentalization. This structural characteristic is suggested to contribute to more stable and potentially efficient symbiotic interactions (Parniske 2018; Faria et al. 2022; Mohd-Radzman and Drapek 2023; Casaes et al. 2023).

2.2 Plant setting and manipulation of bacteria association

To assess the extent to which plants depended on N-fixing bacteria and how much plants invested in root nodulation, we experimentally manipulated the association between *C. nictitans* plants and rhizobia bacteria. To set plants for the experiment, we collected seeds from 15 distinct mother plants found in a natural population at the Federal University of ABC in São Bernardo do Campo city, Brazil. To break the mechanical dormancy of the seeds used in the experiment, we manually rubbed them onto a rough surface. We seated three seeds from the same mother plant into 8 L pots filled with a soil mixture consisting primarily of sand (70% sand mixed with 30% topsoil brand Terra Vegetal ABC Garden), simulating the soil where *C. nictitans* populations occur (see Chinarelli et al. 2022). The nitrogen concentration of this mixed soil was 1.5 g of N per 1000 g of soil, more than three times the average N concentration of agricultural soils (Brady and Weil 2008). After seed emergence, we also applied liquid fertilizer containing nitrogen, potassium, and phosphorus nutrients (see details below). To avoid unintended bacterial infections, we sterilized the soil by autoclaving it before filling the pots with the soil mixture. We sterilized seeds and pots using 70% ethanol, 1% hypochlorite, and 3% hydrogen peroxide and cleaned them with distilled water before seed setting (Somasegaran and Hoben 1994). We followed seed germination and seedling establishment over 90 days. After that, we kept one seedling per pot. We have grown 180 plants separated into two groups: plants inoculated with rhizobia (control R+, $n=90$) and plants non-inoculated with rhizobia (treatment R-, $n=90$). Both experimental groups included seeds from all 15 mother plants. We conducted the entire experiment in a Van Der Hoven® greenhouse at the Federal University of ABC.

We inoculated the R+plant group with an aqueous solution containing rhizobia bacteria. We prepared the solution by grinding root nodules from adult spare *C. nictitans* plants

and diluting them into distilled water. We applied 100 mL of bacteria solution on the soil of each R+plant, first at the seed setting and then one week later. In the R- plants, we applied 100 mL of distilled water to the soil during seed setting and one week later (Godschalx et al. 2015). To avoid cross-contamination between the experimental groups, we sterilized the greenhouse benches where the plants were kept weekly, using 70% ethanol and 3% hydrogen peroxide throughout the experiment.

Starting four weeks after seed emergence, we initiated the application of a liquid fertilizer containing nitrogen, potassium, and phosphorus nutrients. Following the brand's instructions, we used Sempre Verde® N-P-K (10-10-10) fertilizer. The fertilizer applied had a combination of ammonium nitrate (NH_4NO_3) and urea ($(NH_2)_2CO$). Ammonium nitrate is highly soluble, providing both ammonium ions (NH_4^+) and nitrate (NO_3^-) as nitrogen sources readily available for plant uptake. Urea, on the other hand, decomposes in the soil, releasing ammonium (NH_4^+) and various nitrogen oxides, including nitrate (NO_3^-), through microbial activity. This combination ensures that plants in our experiments had access to multiple forms of nitrogen. A small capful of the fertilizer was diluted in 1 L of water and carefully sprayed onto the soil surrounding the base of each plant's stem. Over three months, we applied an average of 35 ml of the diluted solution six times every 15 days. This procedure resulted in a total addition of 180 mg of N during the initial months of plant growth. Substrate preparation and fertilizer application were implemented in both plant treatment groups (R+and R- plants). The combination of available nitrogen in the substrate (organic soil mixture) and the liquid fertilizer application ensured that plants in our experiments had access to multiple forms of nitrogen and did not experience limitations due to nitrogen scarcity in the soil.

The abovementioned procedures also addressed potential constraints caused by phosphorus levels, which might decrease plant growth and nitrogen fixation in R+plants (Pérez-Fernández et al. 2019). Additionally, we applied the Forth ® fungicide on the leaves of all plants to avoid fungi infections. Fertilizer and fungicide application occurred at separate 21-day turns four months after seed germination.

To assess whether the nitrogen provided by the fertilizer was sufficient to support plant growth under our experimental conditions, we conducted an additional experiment using maize (*Zea mays* L.). Maize was selected due to its well-documented rapid growth and high responsiveness to soil nutrients, making it a reliable system for evaluating fertilizer effectiveness. The experiment was conducted under the same conditions established for *C. nictitans*, including the use of a sterilized substrate, a controlled greenhouse environment, and the same fertilizer regimen. Maize plants were cultivated under three experimental conditions: (1) with fertilizer but without rhizobia inoculation (R- equivalent), (2) with fertilizer and rhizobia inoculation (R+equivalent), and (3) without fertilizer but with rhizobia inoculation. Further details of the experiment and results are provided in the supplementary material.

2.3 Measuring plant fitness descriptors

We assessed plant fitness characteristics at two different time points. Initially, we monitored plant survival by census seedlings that perished within the first three months after seed germination. The high mortality rate observed among R- plants during this period (see Results section) led us to evaluate the other fitness descriptors using sub-samples of 10 plants from each experimental group (total $n=20$). These sub-samples were composed of plants originating from the same ten mother plants. Over ten months, we periodically measured plant growth parameters: plant height and leaf count. Additionally, we examined plant reproductive effort by quantifying the number of flower buds and open flowers produced by plants from each treatment group.

Following the assessment of fitness descriptors, we carefully removed all 20 sampled plants from their soil pots, segregating their leaves, stems, and roots into distinct paper bags. Subsequently, we initiated the drying process of all plant material at 50 °C to facilitate the measurement of dry weight. The duration of the drying process ranged from two to four days, depending on the specific plant organ (according to Pérez-Harguindeguy et al. 2013). For R+plants, we determined their investment in N-fixing root nodules by quantifying the number of root nodules and measuring the dry biomass of these nodules (see Fig. 1b).

2.4 Identification of the nitrogen-fixing bacteria in the root nodules of Chamaecrista nictitans

To identify the bacterial species involved in the mutualistic relationship with *Chamaecrista nictitans*, we collected root nodules from the root system of five different plants grown under the same conditions as in the experiment. Nodules were sterilized separately following the Howieson and Dilworth (2016) protocol, washed with distilled water, and macerated inside sterilized microtubes. Total DNA was extracted using commercial DNA extraction kits, and the extracted DNA was used to amplify the entire 16 S rRNA gene via polymerase chain reaction (PCR), using the primers 27 F and 1492R to generate a fragment of approximately 1.6 kb.

The amplicons were analyzed on agarose gel, quantified for library preparation using the SQK-LSK114 kit (Oxford Nanopore Technologies), and sequenced on a Flongles Flow Cell (FLO-FLG114) using the MinION platform (Oxford

Nanopore Technologies). After sequencing, the reads were processed through basecalling using Guppy Basecaller (v6.0.7) and filtered for quality control at Q10 with NanoFilt (v2.3.0). Demultiplexing was performed with Porechop (v0.2.4), and the reads were mapped against a 16 S reference database using KMA (v1.4.3). Data analysis was conducted in Python 3.7, allowing for the identification of bacterial species by comparison with a reference database containing over 15,000 bacterial sequences.

2.5 Data analysis

We applied a generalized linear mixed model (LMM) to assess the effect of N-fixing bacteria on seedling survival after three months of germination. We considered the proportion of surviving seedlings as a response variable, rhizobia inoculation as a fixed factor with two levels, R- and R+, and the mother plants as a random factor.

To assess the impact of N-fixing bacteria on plant height (in cm), we employed LMM. We used plant height as the continuous response variable, considering rhizobia inoculation as a fixed factor with two levels, R- and R+, and the mother plant as a random factor. To investigate the effect of N-fixing bacteria on the total number of leaves we conducted a Generalized Linear Mixed Model (GLMM) using a negative binomial distribution. In this model, the total number of leaves was the response variable, rhizobia inoculation was the fixed factor with two levels, R- and R+, and the mother plant was the random factor.

To evaluate the impact of N-fixing bacteria on plant reproductive investment, we applied a GLMM with a negative binomial distribution. We considered the total sum of buds and open flowers as the response variable, rhizobia inoculation as a fixed factor with two levels, R- and R+, and the mother plant as the random variable.

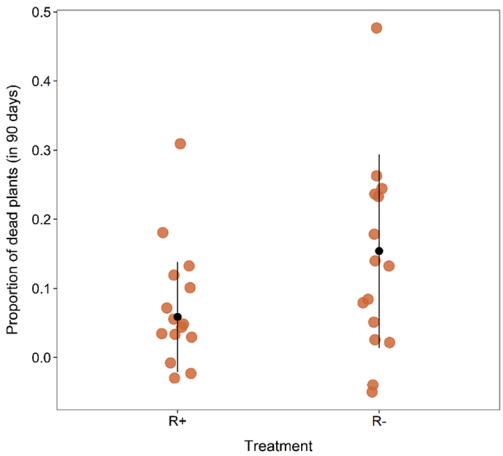


Fig. 2 The relationship between plant height and the production of floral buds and flowers in *Chamaecrista nictitans* plants inoculated (R+) and non-inoculated (R-) with rhizobia. (**a**) Plant height (cm) at 4 and 12 months of plant age. (**b**) Production of flower buds and flowers after 12 months of plant growth. The points are the mean number of surviving individuals for each mother plant at the end of the study

To investigate the influence of N-fixing bacteria on the root dry mass (in grams) of both R+and R- plants, we conducted an LMM test, with root dry mass as the response variable and the presence and absence of rhizobia as the fixed factor. To assess the relationship between the investment in N-fixing root nodules and the total dry biomass, we applied an LMM. We considered the plant's dry weight (in grams) of R+plants as the response variable and the total dry weight of root nodules as the predictor variable. The plant's dry weight comprised the leaf, stem, and root dry weights. In both analyses, we used the mother plant as the random variable.

In all models, we assessed the homogeneity of variance and normality of residuals using graphic visual inspections. We performed all analyses in the R environment 4.2.3 (R Development Core R Core Team 2023).

3 Results

3.1 Plant growth and survival

Within the first three months following seed germination, the mortality rate was 2.5 times higher in non-inoculated plants (R-) compared to inoculated plants (R+) (LMM: df=14, F=9.3, p-value=0.008; Fig. 2). Survivor R- plants exhibited a clear pattern of leaf chlorosis (Fig. 1C), a symptom of nutrient (especially N) deficiency. We also observed differences in plant growth between R- and R+plants. By the third month, it became evident that R- plants were smaller than their R+counterparts (Fig. 1c-d). At the 390-day mark, R- plants were, on average, 8.48 cm shorter in height (cm) compared to R+plants (LMM: n=20 plants; df=18, F=60.5, p-value<0.001; Fig. 3a). On average, the R- plants produced 9.21 fewer leaves than R+plants (GLMM negative binomial: n=20 plants; df=16, z=8.8; p-value<0.001; Fig. 3b).

3.2 Reproductive output

The R- plants did not produce any flower buds or flowers throughout the experiment. In contrast, the R+plants yielded 34.9±33.6 (mean±SD) floral buds and 6.0±5.7 flowers (GLMM negative binomial for buds+flowers: df=1, F=3.83, p-value=0.0003; Fig. 4). After the experiment, R- plants had no root nodules, confirming the effectiveness of our sterilization procedure. In contrast, R+plants consistently developed root nodules (Fig. S3). Notably, the number of root nodules per plant exhibited wide variability, ranging from 14 to 517 root nodules per plant.

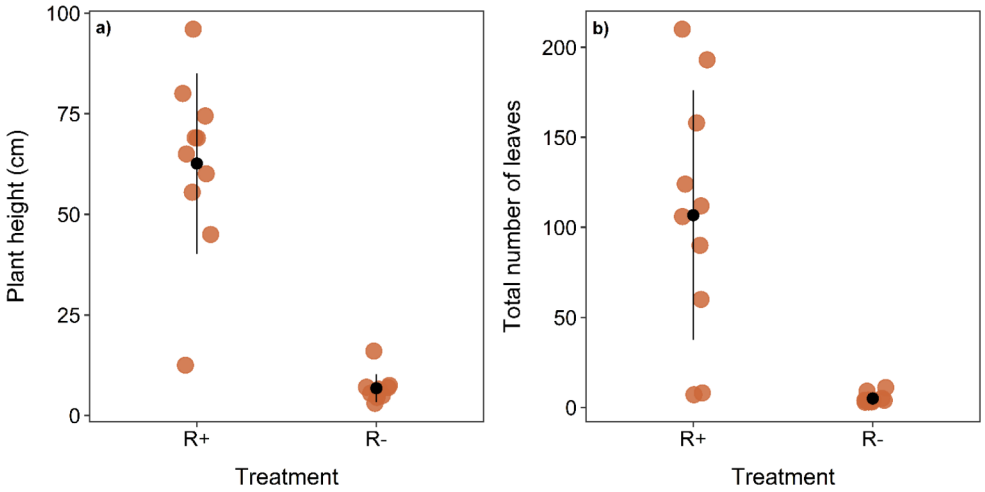


Fig. 3 Plot showing the variation in the relative proportion of mortality of *Chamaecrista nictitans* plants due to the presence or absence of nitrogen-fixing rhizobia during the first three months of plant growth. Non-inoculated (R-) plants are more likely to die than inoculated (R+) plants

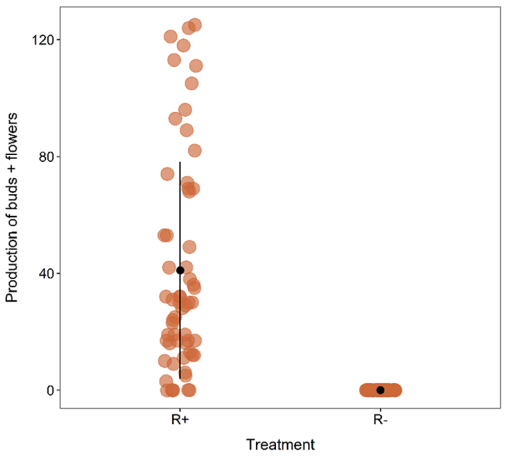


Fig. 4 Number of root nodules produced by inoculated (R+) and non-inoculated (R-) *Chamaecrista nictitans* plants. R- plants did not form nodules

The dry root mass (g) differed among plants from the two treatments, with R- plants having lower dry root mass than R+plants (LMM: df=18, F=20.8, p-value=0.0002; Fig. 5a). Regarding R+plants, we observed a significant positive relationship between the weight of root nodules and the total plant biomass (g) (LMM: df=18; F=47.1, p-value<0.001; Fig. 5b). On average, each 0.1 g increase in root nodule weight was associated with a 3 g increase in total plant biomass.

3.3 Identification of nitrogen-fixing bacteria from root nodules

The bacteria found in 1 to 6 nodules per plant across five different *C. nictitans* plants clustered into a single group based on 16 S rRNA gene profiles, all identified as belonging to the genus *Bradyrhizobium*.

4 Discussion

We observed that *C. nictitans* individuals not associated with N-fixing bacteria (R-) faced a higher mortality than those associated with them. Additionally, R- plants that survived after three months of germination were much smaller and exhibited stronger signs of nutritional deficiency than

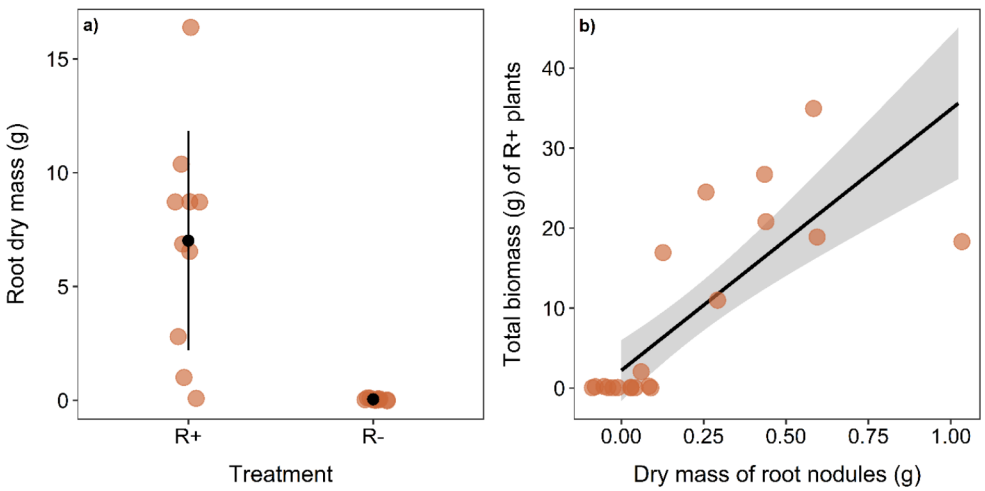


Fig. 5 (**a**) Positive relationship between plant dry biomass, including root dry biomass without nodules (R-), and (**b**) root nodules dry biomass individuals of *Chamaecrista nictitans* inoculated with bacteria (R+)

bacteria-inoculated plants (R+). R- plants also produced no flower buds, indicating that the association with nitrogen-fixing bacteria is fundamental to the reproduction of *C. nictitans*. Considering that R+and R- plants relied on a high constant supply of free N in the soil during our experiments, these negative effects of non-inoculation indicate that *C. nictitans* strongly rely on the N fixed by their partner bacteria, even when alternative sources of N are available. Our parallel experiment with *Zea mays* also supports this conclusion by demonstrating that the fertilizer used in our study is sufficient to sustain healthy plant growth. Altogether, these results indicate that *C. nictitans* maintains is highly dependent on *Bradyrhizobium* N-fixing bacteria.

Mutualistic interactions are expected to confer fitness benefits to all involved partners (Althouff & Segraves, 2016), and such effect should be even more substantial in obligatory interactions (Chomicki et al. 2020). The positive outcomes observed in the R+plants are likely due to the various beneficial mechanisms associated with the presence of N-fixing bacteria in the root system of *C. nictitans*. R+plants received a consistent supply of N from rhizobia, whereas R- plants relied solely on soil N, leading to increased mortality rates, reduced growth, and suppressed reproduction. The association with rhizobia provides R+plants with a beneficial alternative N source, a limiting nutrient for plants (Ågren et al. 2012), allowing for more efficient N acquisition, provided that the benefits of this mutualistic relationship outweigh its costs (Bronstein 2001), enhancing plant growth, survival, and reproduction (Walker et al. 2012; Lindström and Mousavi 2020; Mathesius 2022). Additional evidence supporting the idea that *C. nictitans* heavily rely on *Bradyrhizobium* N supply came from pilot experiments performed by our research group. We sprayed a rhizobia solution onto the soil of some non-inoculated *C. nictitans* plants that had survived as undeveloped seedlings for nearly a year. The root infection by rhizobia, even one year after germination, enabled the plants to grow and develop normally. This suggests that N-fixing bacteria are crucial for the growth and reproduction of *C. nicitans*, and that the plants can maintain their growth potential until root infection occurs. It is important to consider, though, the presence of N-fixing bacteria can influence plant physiology beyond N-fixation. Certain rhizobia species produce plant growth-promoting hormones during root nodule formation, which are especially valuable under N-deficit conditions. For example, cytokinins and auxins play crucial roles in helping plants cope with N and other nutrient limitations. Cytokinins stimulate the formation of root nodules, thereby enhancing N-fixation efficiency (Nagata and Suzuki 2014). They also boost nitrate reductase activity, improving N metabolism and the plant's ability to utilize available nitrates in the soil (Nagata and Suzuki 2014). Auxins facilitate rhizobia infection and subsequent nodule formation, optimizing the absorption of N and other essential nutrients (Patil et al. 2017). These hormones enable plants to better manage N deficiency by promoting root development, delaying senescence, and maintaining overall vigor (Lin et al., 2020). These joint beneficial effects of the association increase as the number of root nodules increases, as demonstrated by the fact that R+*C. nictitans* plants producing more root nodules were those accumulating greater biomass.

Besides plant survival and growth, the mutualistic interaction with rhizobia also positively affected the flower production of *C. nictitan*s, and plants without mutualism did not produce any flowers. Some N-fixing bacteria produce gibberellin (Nett et al. 2022), a plant hormone that can enhance flower bud and seed production (Dobbelaere et al. 2003). Thus, the plant association with rhizobia in *C. nictitans* may represent direct and indirect advantages for plant reproduction, favoring the persistence of these plants in the environment and shaping the higher plant dependency on the interaction over evolutionary time. This substantial effect of nutritional mutualism with N-fixing bacteria on plant reproduction has been detected before, including a species of the genus *Chamaecrista*. In Souza et al. (2024), *Chamaecrista latistipula* plants inoculated with N-fixing bacteria in N-poor soil were much larger and more attractive to pollinators than non-inoculated plants growing in N-rich organic soil. The non-inoculated plants produced only a few flowers during the experiment (Souza et al. 2024). The responses of *C. nictitans* and *C. latistipula* to the experimental exclusion of rhizobia partners show that something very particular has occurred, at least in some lineages within the *Chamaecrista* genus, increasing the plants' dependence on nitrogen-fixing bacteria. Increasing dependence may culminate in the total or partial loss of plant's ability in obtaining N from the soil, perhaps because they are associated with bacteria that are efficient in fixation and are widespread in natural systems.

The strong mutualism exhibited by *C. nictitans* (and *C. latistipula*, see Souza et al. 2024) with N-fixing bacteria differs from other bacteria-legume associations, which are mostly facultative (Maróti and Kondorosi 2014). This high dependency is rare and is typically observed in highly specialized systems (Blüthgen et al. 2007; Smith and Leebens-Mack 2024). When mutualistic partners differ in quality, natural selection promotes mechanisms that enable mutualists to select their partners (Bronstein et al. 2001). This mechanism is particularly advantageous when high-quality partners are relatively common in the environments. This selection process leads to specialization, often correlated with higher dependency (Chomiski et al. 2020). Our analysis using molecular markers revealed that only bacteria from the *Bradyrhizobium* genus were present in the root nodules of *C. nictitans*. *Bradyrhizobium* is a widely distributed and

dominant genus found in many regions across the globe, especially in tropical and subtropical areas (Banasiewicz et al. 2021 and references therein). *Chamaecrista nictitans* rely heavily on bacteria for N acquisition, which is likely selected for N-poor and degraded environments. Notably, *C. nictitans* individuals are commonly found in disturbed habitats (Space et al. 2004), and their strong dependence and specificity on N-fixing *Bradyrhizobium* may confer certain advantages. Unlike species with strictly facultative strategies, which must produce and deactivate nodules or trigger and inactivate N-fixing bacteria, resulting in greater energy costs, plants with a high dependency on N-fixing bacteria maintain a constant investment in nodules (Menge et al. 2009). Maintaining a stable N-fixing capacity with a predictable and advantageous partner, *C. nictitans* ensures a reliable N supply for growth and development, even under variable environmental conditions (Menge et al. 2009). Although variability in rhizobia efficiency across and within plants is well-documented (Heath and Tiffin 2009; Batstone et al. 2017), this variation is expected to be lower in *C. nictitans*. The relative efficacy of rhizobia as nitrogen providers likely interacts with soil traits (e.g., N content), but this interaction was not observed in our study. The high dependence on N obtained from symbionts, associated with the ubiquity of *Bradyrhizobium* bacteria in the soil, may contribute to *C. nictitans* persistence and success in disturbed ecosystems. This annual plant optimizes growth and reproduction associated with these bacteria in the root system, ensuring reliable access to essential nutritional resources within its short lifespan. It is important to note that we have not assessed the impact of plants on these bacteria. Mutualistic relationships are asymmetric, with one partner more dependent than the other (Chomicki et al. 2020). Thus, the high plant dependence on bacteria does not imply a reciprocal dependency. Instead, bacteria may interact facultatively, remaining freely available in the soil and ready to associate when plant seeds germinate — an aspect warranting further investigation.

The high dependency of *C. nictitans* on nodulation can also impact other plant species in the surrounding environment. When *C. nictitans* plants shed their leaves or die annually, they increase soil N availability for surrounding plants (see Fornara and Tilman 2009). Interestingly, as *C. nictitans* is highly dependent on association with bacteria, subsequent generations cannot use the N incorporated in the soil by previous generations (Menge & Hedin, 2009). Consequently, the soil N resulting from the annual accumulation of *C. nictitans* biomass is utilized by neighboring plant species rather than *C. nictitans* offspring (see Ren et al. 2008). Thus, it is likely that *C. nictitans* acts as a facilitator plant species by enhancing soil N availability to other plant species over time (Padilla and Pugnaire 2006), probably acting as a keystone species during ecological succession in disturbed areas (Shumway 2000; Su et al. 2002; Cathey et al. 2010; Yelenik et al. 2015, Venier et al. 2023).

Considering that previous studies described facultative associations between *Chamaecrista* species and N-fixing bacteria (but see Souza et al. 2024), the high dependence of *C. nictitans* on their N-fixing partners highlights that obligatory and facultative interaction with N-fixing bacteria can coexist within *Chamaecrista* genus. This variation has likely been shaped by the geographical distribution of *Chamaecrista* species and life strategies (annual or perennial). By understanding the level of dependence of plants on their mutualistic bacteria, we can better predict how changes in these interactions may affect the overall structure and dynamics of ecosystems in which they are embedded. Due to their effect on surrounding plants, our results place *C. nictitans* as a potential species for developing sustainable agricultural practices that preserve the natural nutrients supply in the soil. Furthermore, we reinforce empirical studies' critical significance in understanding the underlying mechanisms that govern the nature and extent of dependence between mutualistic partners in tropical ecosystems.

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Author contributions The study was designed by Caroline Souza, Anselmo Nogueira, and Laura C. Leal. Caroline Souza, Bruno Melati, and Anselmo Nogueira performed the experiment setup with the legume species. Data collection was carried out by Caroline Souza, Anselmo Nogueira, and Vanessa Lima. Caroline Souza and Anselmo Nogueira performed analyses. Anselmo Nogueira and Alessandro L. Urzedo performed the experiment setup with the maize. Laura C. Leal and Anselmo Nogueira provided funds. Caroline Souza wrote the first draft of the manuscript, and all authors commented on previous versions. All authors helped with data interpretation, writing the article, and approving the final manuscript.

Declarations

Competing interests The authors declare no competing interests

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